EXP 7

Data Visualization Using Power BI

AIM:

To learn the Power BI Desktop workflow by connecting to Excel and CSV data, cleaning the data in Power Query, creating basic visualizations (bar, line and pie charts, cards and tables) and arranging them as an interactive dashboard.

ALGORITHM:

1. Open Power BI Desktop and identify the Fields, Visualizations and Filters panes, along with the Report, Data and Model views.
2. Click Home → Get data → Text/CSV, select the employee attrition dataset and click Load.
3. Click Transform data to open Power Query Editor and clean the dataset by handling missing values, removing duplicates and correcting data types.
4. Verify that columns such as Department, Job Role, Gender, Monthly Income, Job Satisfaction and Attrition have the appropriate data types.
5. Click Close & Apply to load the cleaned employee data into Power BI.
6. Create four Card visuals displaying Total Employees (1,470), Total Attrition (237), Average Monthly Income (approximately 7K) and Average Job Satisfaction Score (3).
7. Create a clustered bar chart showing attrition by job role and distance from home, comparing employees who stayed and those who left.
8. Create a clustered bar chart to compare average monthly income by education level and attrition status.
9. Create a column chart showing job satisfaction scores and their corresponding attrition percentages.
10. Create a pie chart for promotion history, a treemap for department-wise employee distribution and a stacked bar chart for gender-wise attrition.
11. Arrange all KPI cards and charts neatly on one dashboard page, apply consistent colours, titles, legends and data labels, and save the report as a .pbix file.

OUTPUT



RESULT:

Thus, the Power BI Desktop workflow was studied. Data was imported from Excel and CSV sources, cleaned using Power Query, and visualized using bar, line and pie charts, cards and tables, and an interactive dashboard was prepared.